

Fractal Geometry

Evaluation Policy :

1. Mid-term Exam — 30%
2. End-term Exam — 40%
3. Quizzes (2) — 15% each

Instructor:

Amit Priyadarshi

Email : priyadarshi@maths.iitd.ac.in

Office : 527 E, 99 B (Academic Building West)

Audit Pass : Minimum 40%

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Reference Books:

1. K. J. Falconer, Fractal Geometry: Mathematical Foundations & Applications, Wiley
2. Gerald Edgar, Measure, Topology and Fractal Geometry, Springer
3. M. F. Barnsley, Fractals Everywhere, Dover.

• Basic Measure Theory

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What are fractals?

1. Cantor Set

$$E_0 = [0, 1]$$

$$E_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$$

$$E_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$$

$\vdots$
 $E_n =$ union of 2^n closed intervals
 each of length $\frac{1}{3^n}$.

Cantor set, $E = \bigcap_{n=0}^{\infty} E_n$



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Some properties:

1. E is an uncountable set.
2. "Lebesgue measure" of E is zero.

$$\begin{aligned} m([0, 1] \setminus E) &= \frac{1}{3} + 2 \times \frac{1}{3^2} + 2^2 \times \frac{1}{3^3} + \dots \\ &= \frac{1}{3} \left(1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots \right) \\ &= \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = 1 \end{aligned}$$

$$\Rightarrow m(E) = 1 - 1 = 0.$$

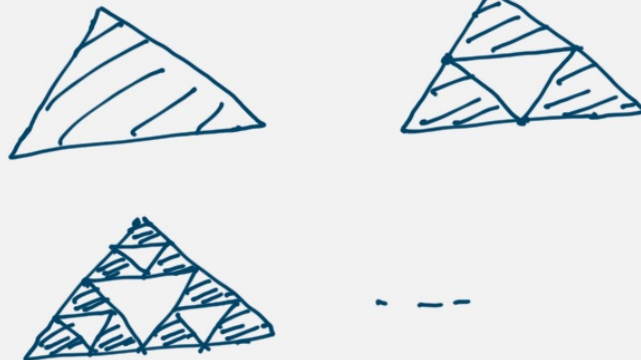
3. "Self-similarity"

$$E = \underbrace{(E \cap [0, \frac{1}{3}])}_{\text{scaling factor} = \frac{1}{3}} \cup \underbrace{(E \cap [\frac{2}{3}, 1])}_{\text{are similar to } E}$$

scaling factor = $\frac{1}{3}$

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Sierpinski gasket

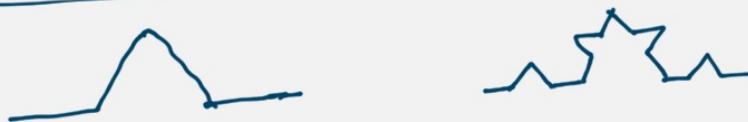


- Uncountable
- Zero Lebesgue measure
- Self-similar with scaling factor = $\frac{1}{2}$

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Koch Snowflake curve



and so on.

This is also self-similar with
scaling factor = $\frac{1}{3}$

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Basic Measure Theory

Outer measure

Let X be any set.

An outer measure μ on X is

$$\mu: \mathcal{P}(X) \rightarrow [0, \infty]$$

($\mathcal{P}(X)$ = the power set of X
= set of all subsets of X)

satisfying

- (i) $\mu(\emptyset) = 0$
- (ii) $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$ (Monotonicity)
- (iii) $\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$ (Countable subadditivity)

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Examples :

1. Counting measure

$$\mu: \mathcal{P}(X) \rightarrow [0, \infty]$$

$$\mu(A) = \begin{cases} \text{no. of elements in } A & , \text{ if } A \text{ is finite} \\ \infty & , \text{ if } A \text{ is infinite} \end{cases}$$

2. Lebesgue ^{outer} measure on $\mathbb{R}$

$$\mu: \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$$

$$\mu(A) = \inf \left\{ \sum_{j=1}^{\infty} \ell(I_j) : A \subset \bigcup_{j=1}^{\infty} I_j, \right. \\ \left. I_j \text{ is an interval} \right\}$$

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